Predicting Student Dropout, with a Fairness Audit

Four mistakes. Each one changed an answer.

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PGD AI/ML capstone, Pillar 5

UCI, Predict Students' Dropout and Academic Success. 4,424 students, Instituto Politecnico de Portalegre

The frame

3,630
students, after 794 with no settled outcome come out

39.1%
drop out. Mildly uneven, handled with class weights

12
curricular columns dropped as leakage

The task is binary classification, Dropout against Graduate. Enrolled means the outcome is not settled, so those rows cannot teach the model what dropout looks like.

The twelve curricular columns record first and second semester performance. They predict dropout almost perfectly, because a student failing courses is a student already leaving. The model sees only what is known at enrollment.

PR AUC and ROC AUC choose the model, because they hold across every cutoff.

Recall judges the deployed system, and only once a cutoff exists. Recall is not a property of a model. It is a property of a model and a cutoff.

Accuracy is reported and not trusted. A model that calls everyone a graduate is right 61 percent of the time and catches nobody.

The gaps were there before the model

Dropout rate by group, against an overall 39.1 percent. Notebook 02.

Group	Dropout rate	n
Men	56.1%	1,249
Women	30.2%	2,381
No scholarship	48.4%	2,661
Scholarship holder	13.8%	969
Debtor	75.5%	413
Not a debtor	34.5%	3,217
Age 24 to 30	66.5%	499
Age 17 to 20	26.1%	2,080

94.0%

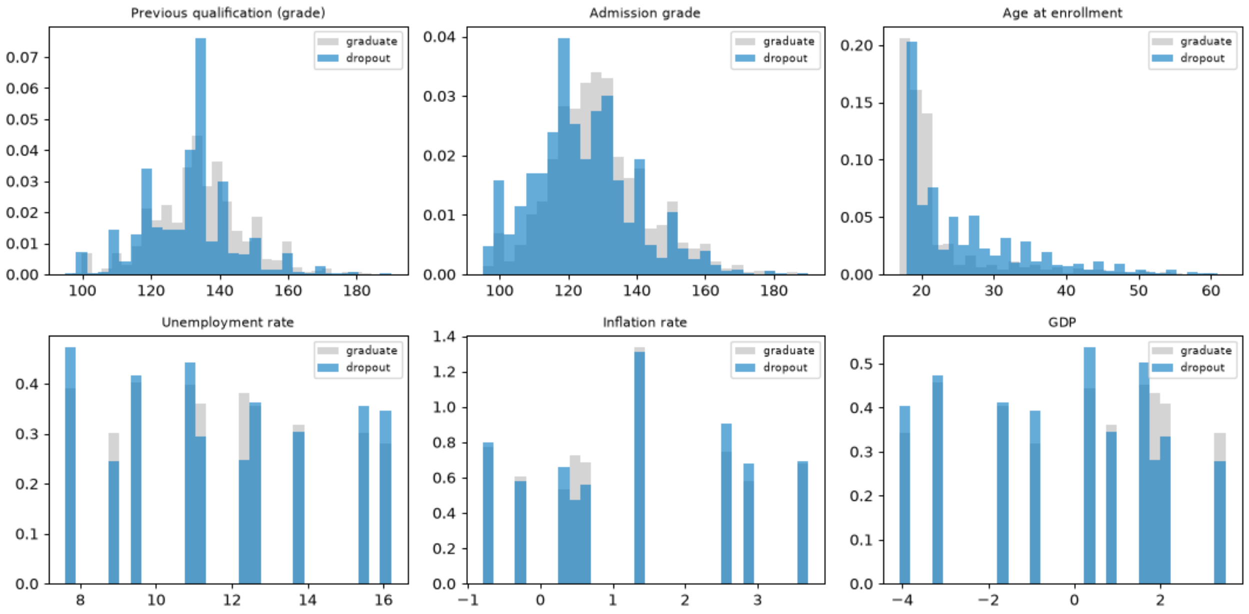
Dropout rate for the 486 students not up to date on tuition, against 30.7 percent for those who are.

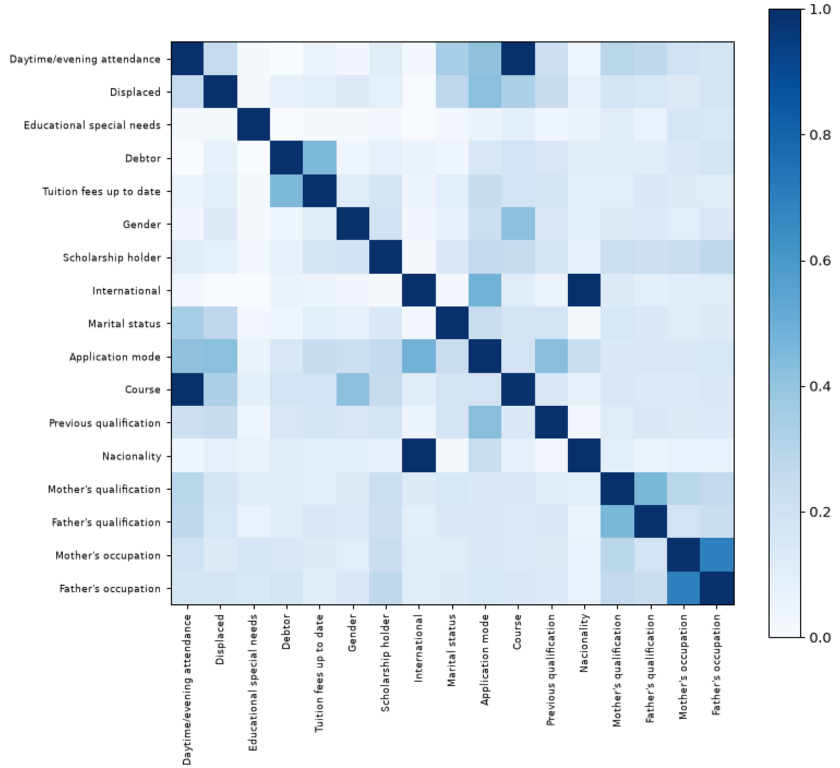
That is not a background fact. A student who has stopped paying is often a student already leaving. It is a near-outcome signal, kept but watched, and Step 3 measures what the model loses without it.

These numbers come back on slide 10. They are the reason the fairness audit had to be rewritten.

What predicts dropout

The right test for each feature type. Mann-Whitney U on the numbers, chi-square and Cramer's V on the categories.





Age has the strongest link, p = 9.2e-65. Older entrants drop out more. Admission grade and previous qualification grade follow. Unemployment and inflation show no real difference, p = 0.75 and 0.21.

Tuition 0.437, Course 0.346, application mode 0.330 lead on Cramer's V.

Every VIF is under 2. But VIF only ever looks at the six numerics.

The matrix shows two pairs sitting at a perfect V = 1.000. I noted that and moved on.

1 The feature that broke the coefficients

A socioeconomic pressure score, one point each for owing money, falling behind on tuition, and holding no scholarship. It ranked first on mutual information, ahead of every raw column in the file.

Then the model said owing the school money makes a student less likely to drop out.

```
pressure − Debtor + Tuition + Scholarship = [2]
```

One value. Exactly 2, on all 2,904 training rows. The score was three columns the matrix already held.

Design matrix	Old	Fixed
Columns	216	84
One-hot blocks	9	6
Exact linear dependencies	19	6

Slide 4 said collinearity was fine. VIF said so. VIF only reads the six numerics. It never saw the one-hot columns, and it never saw a feature built after it ran.

Same model, same data. The only change is whether the score sits in the matrix.

Coefficient	Score in	Score out	Real rate
Debtor	−0.465	+1.006	76% vs 34%
Scholarship holder	+0.138	−1.344	14% vs 49%
Tuition fees up to date	−1.480	−2.913	31% vs 94%

Two signs flip. The model reads owing money as protective and a scholarship as a risk. The data says the opposite, loudly.

Accuracy never moved. The score carried no information the model did not already have. What it did was make the model unreadable, and every explanation in Step 5 is read off these coefficients.

A feature that adds no information cannot help a model, but it can destroy its ability to explain itself.

Features that replace, instead of repeat

A feature built from columns that stay in the matrix adds nothing to a linear model. It earns its place only by replacing its sources.

The same rank check caught two more repeats, both already flagged at V = 1.000 and both waved past.

International equals (Nacionality != 1) on 100 percent of rows. The same column twice.

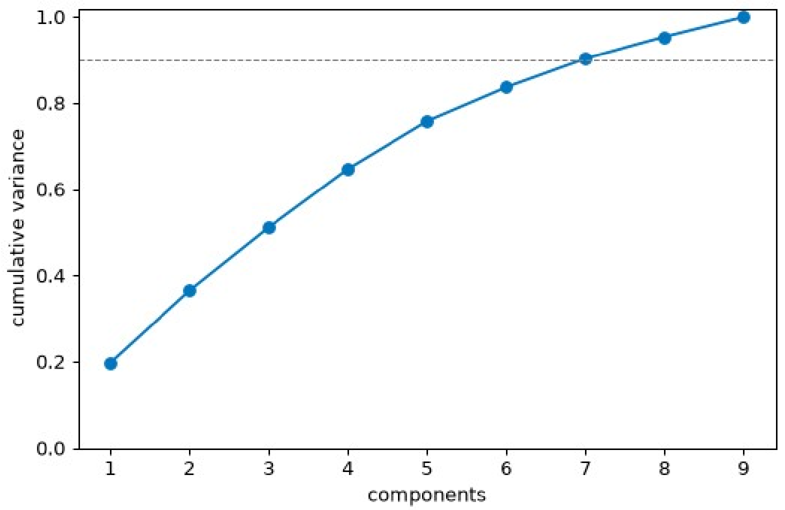
Daytime attendance is decided entirely by Course. Zero of 17 courses run both a day and an evening class.

Five engineered features replace their sources. Four weak columns come out. Then the cost gets measured, on training folds, rather than assumed.

mature entry lands sixth on mutual information at 0.0534, above every raw column it was built from.

Feature set	CV PR-AUC	Columns
Every raw feature	0.817	215
Engineered and selected	**0.819**	**84**

215, not the 216 on the last slide. The difference is the pressure score, now gone.



PCA takes seven of nine components to reach 90 percent of the variance. Nothing to compress, which matches the low VIF. So PCA rides into Step 4 as a seventh model rather than sitting in a chart nobody consumes.

Sixty percent of the columns gone, and the score went slightly up. Cutting 131 columns for free is the result.

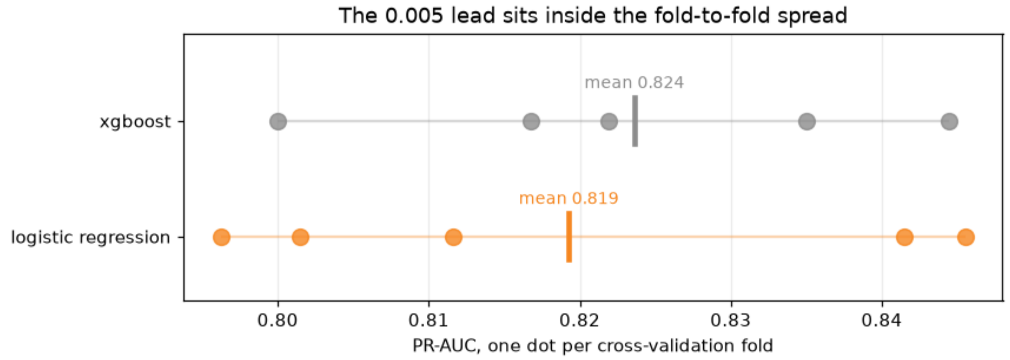
Seven models, and whether the lead is real

Tuned inside five-fold stratified cross-validation on the training set. The test set is scored once.

Model	CV PR-AUC	Test PR-AUC	ROC-AUC
XGBoost	0.824	0.828	0.857
Random forest	0.820	0.823	0.855
Logistic regression	**0.819**	**0.822**	**0.851**
Logistic regression + PCA	0.818	0.821	0.852
SVM	0.808	0.817	0.858
MLP (neural net)	0.752	0.753	0.798
Decision tree	0.723	0.699	0.762

Logistic regression ships. Not because it scores highest, because it does not. When two models are tied, the tiebreak should be something that matters, and here that is whether a student can be told why they were flagged.

PCA lands fourth, one thousandth behind. A principal component is a blend of nine columns, so a student flagged by it could never be told why in a sentence that means anything.



The gap is 0.005. The folds swing by 0.020.

Each dot is one fold. The two ranges overlap almost completely, and the distance between the means disappears inside them.

XGBoost is not better than logistic regression here. It is tied with it, and it happened to land on top.

A gap smaller than the measured noise is not a gap. This deck has to keep that standard everywhere.

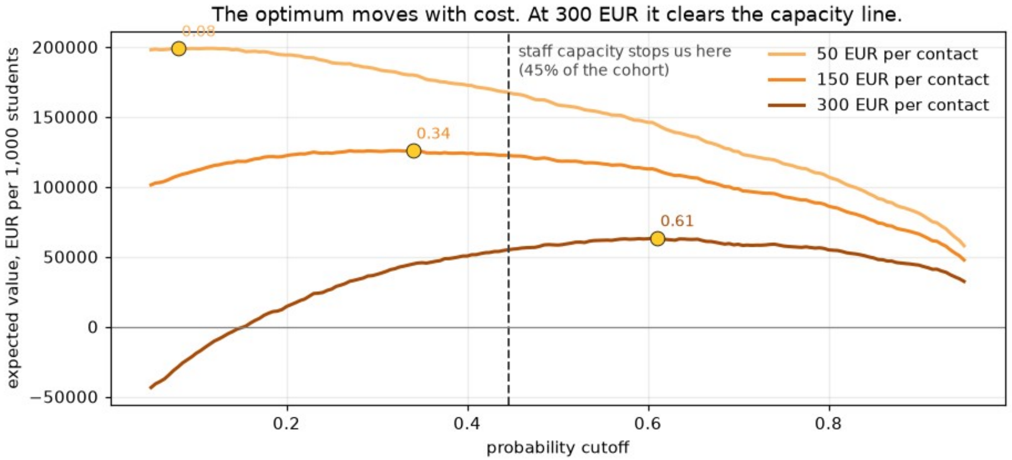
The threshold nobody chose

Recall is not a property of a model. It is a property of a model and a cutoff. The default cutoff is 0.5, and nobody chose it.

Rule	Cutoff	Recall	Precision	Flagged
Default 0.5	0.50	0.750	0.670	43.8%
Capacity 50%	0.38	0.838	0.609	53.9%
Capacity 45%	**0.45**	**0.796**	**0.644**	**48.3%**
Capacity 40%	0.50	0.750	0.670	43.8%
Capacity 35%	0.57	0.697	0.723	37.7%

The default is capacity 40 percent. A staffing decision made by accident, by a library default. Move it to 45 percent and recall climbs to 0.796 on the same model and the same data.

Outreach cost	Best cutoff	Flagged	Value per 1,000
50 EUR	0.08	93.1%	199,022 EUR
150 EUR	0.34	59.4%	125,289 EUR
300 EUR	0.61	34.4%	58,967 EUR



The chart caught me being sloppy, and the table had let me get away with it. I had written that staff time runs out long before the economics do.

The 300 EUR curve peaks at 0.61, tighter than the capacity line.

Staff capacity binds at 50 and 150 EUR per contact. At 300 EUR the money binds first, and the economics want fewer flags than the team could handle.

At the deployed 0.445 the value holds, 171,942 / 123,595 / 51,074 per 1,000. At 150 EUR that is 1,700 below the best case, so capacity is nearly free.

The ranked list is the product. The binary flag is not. The cutoff goes at 0.445, to audit fairness and report one honest number.

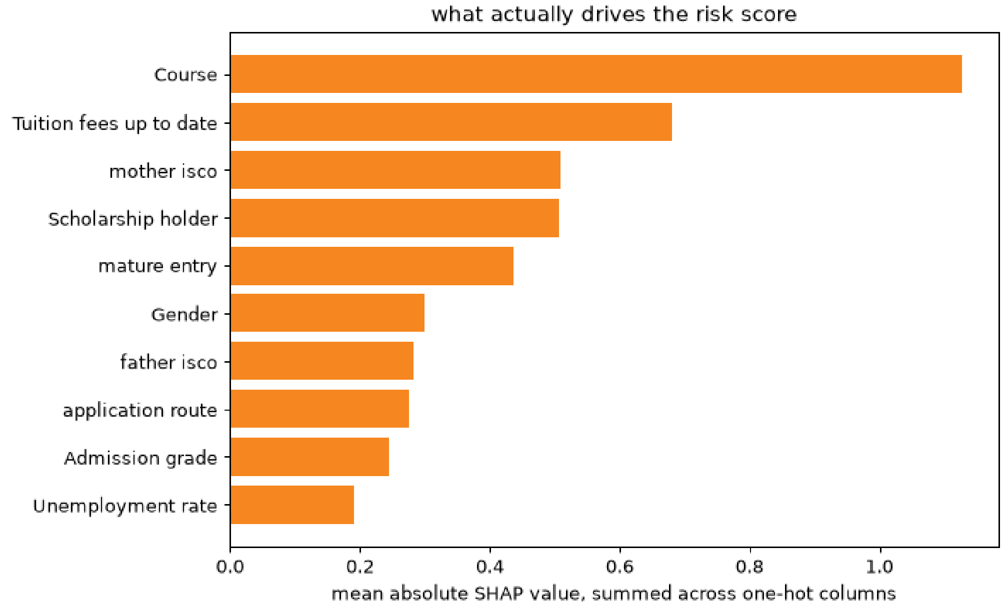
2 SHAP ranked columns, not features

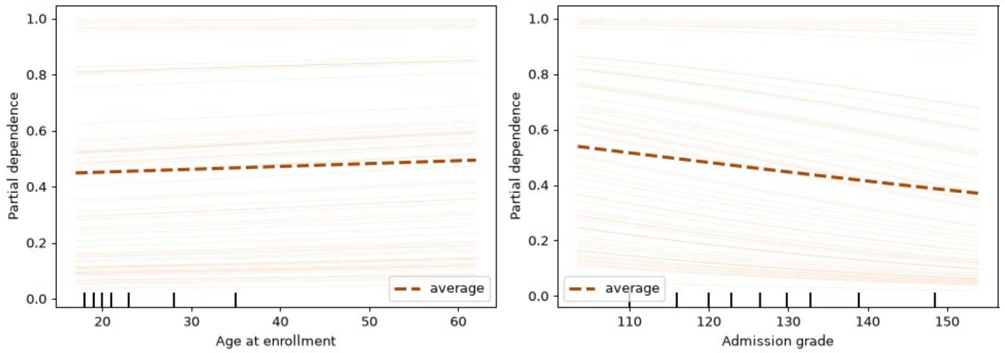
The first chart plotted one bar per column of the design matrix. But Course is not a column. It is seventeen one-hot columns, so its importance got chopped into seventeen small bars while a single flag kept all of its weight in one place.

Old, per encoded column		Fixed, summed by feature	
flag__Tuition fees up to date	0.680	Course	1.126
flag__Scholarship holder	0.507	Tuition fees up to date	0.680
flag__mature entry	0.436	mother isco	0.509
flag__Gender	0.301	Scholarship holder	0.507
cat__Course_9500	**0.284**	**mature entry**	**0.436**
num__Admission grade	0.245	Gender	0.301

cat__Course_9500 is the proof. The largest single fragment of the strongest feature in the model sat fifth, behind three binary flags. Mother's occupation did not appear at all.

Course far ahead at 1.126. Neither a course nor a parent's occupation is a warning a student can act on. The model substantially finds students who started from further back.



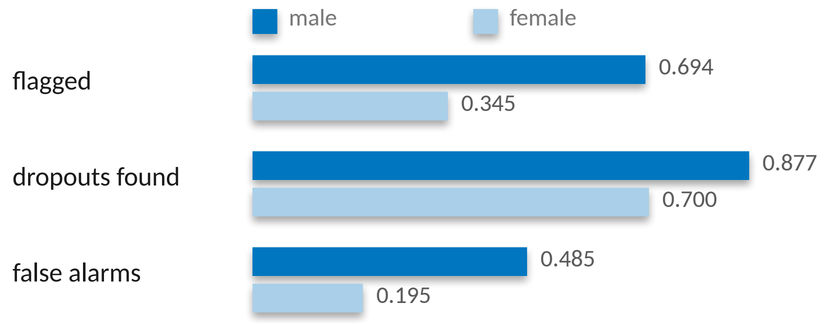


Partial dependence with ICE. The lines run parallel, a check passing rather than a finding.

3 Demographic parity asked the wrong question

Debtors drop out at 76 percent against 34. Demanding the model flag both groups equally is demanding it be wrong on purpose. So the real-world gap goes into the table, next to the metric. If the flagging gap is about the size of the outcome gap, the model reports a disparity. If it is bigger, the model makes one.

Attribute	Real gap	Flagging gap	DI ratio	Error gap	Recall gap
Gender	**0.238**	**0.350**	**0.496**	**0.290**	**0.177**
Scholarship	0.318	0.500	0.171	0.385	0.283
Debtor	0.392	0.369	0.545	0.199	0.199
Age band	0.433	0.558	0.369	0.572	0.253



Debtor. Real gap 0.392, flagging gap 0.369. The model's disparity is smaller than the one in the world. Its DI ratio of 0.545 sits below the 0.8 line, and the old audit called that a failure. It is not.

Gender. Real gap 0.238, flagging gap 0.350. The gap the model creates is bigger than the gap in the world. That is amplification.

3 in 10

women who go on to drop out are never flagged at all. For men it is closer to one in eight. That appears in no flagging-rate metric. It only appears once you stop counting flags and start counting errors.

That is the finding. Not that every attribute fails disparate impact, which was true and useless.

4 Rigor does not generalize on its own

Step 4 refused to call a 0.005 lead real against a 0.020 fold swing. Then I came here and quoted fairness numbers to three decimals off a single 726-row split, without asking what the noise was. So I went back and asked. 2,000 seeded bootstrap resamples.

```
male     n=288   dropouts 154
female   n=438   dropouts 130
```

Women are the majority of the test set and the minority of its dropouts. The recall gap rests on 130 students, not 726.

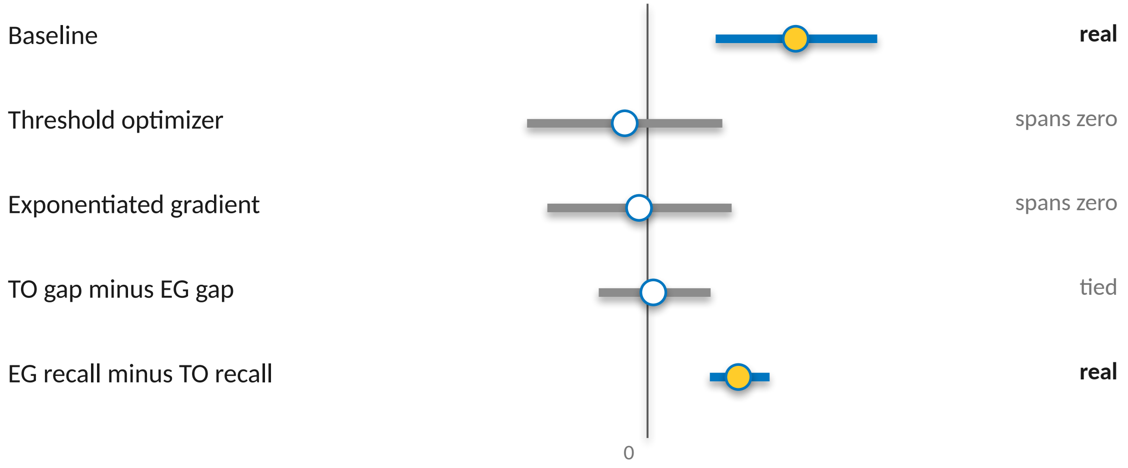
Approach	Recall	Accuracy	Recall gap (magnitude)
Baseline at 0.45	0.796	0.748	0.177
Threshold optimizer	0.609	0.781	0.026
Exponentiated gradient	**0.718**	**0.760**	**0.009**

The harm is real. The 0.177 gap never crosses zero across two thousand resamples.

0.009 is not better than 0.026. They are the same number wearing different clothes.

One difference survives, and it is cost. Eight points of recall against nineteen.

95% intervals, 2,000 resamples. The top three are the recall gap itself, male minus female. The bottom two compare the two mitigations against each other.

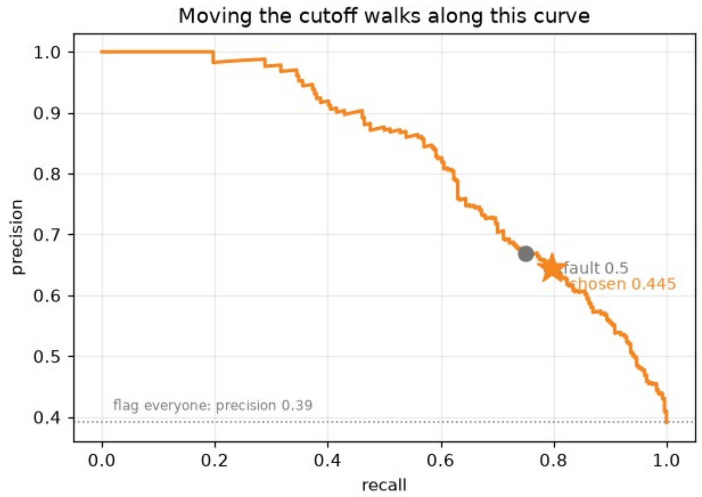


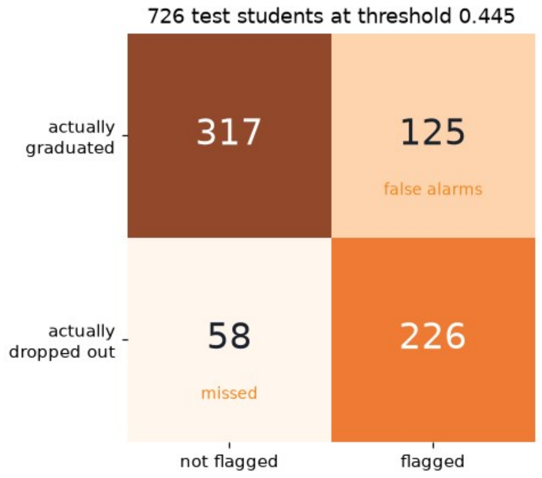
One interval clears zero on the harm, and one clears it on the cost. Everything between them is a tie. The sign flips after mitigation, so the mitigation table reports magnitudes and this one is signed.

The threshold optimizer closed the gap largely by catching fewer people. Accuracy went up while the tool got worse at its job. Fixing a gap by helping fewer people is not a fix.

The verdict, and what it costs

726 test students at the operating threshold of 0.445.





226
leavers caught

58
missed

125
graduates contacted who were never going to leave

Recall 0.796 at a cutoff chosen against staffing, on enrollment-time data only.

Train PR AUC 0.840 against test 0.822, a gap of 0.018. The model learned the pattern, not the noise.

Gender is mitigated. Scholarship and debtor are deliberately not, because those gaps are reported rather than invented. Age band is unresolved, error gap 0.572, and it needs its own pass.

Those 125 are the real cost of this tool. A student pulled into a retention conversation they did not need has been told, in effect, that a system thinks they are failing.

A dropout model is not a scoreboard. It is a decision about which students get help. The right way to use this one is as a ranked prompt for a support team, checked by a person.